

Roll No.

41476

**B. Pharma. 4th Semester
(Regular/Re-appear/Impr.)
Examination – May, 2024**

**PHARMACEUTICAL ORGANIC CHEMISTRY-III
Paper : BP401T**

Time : Three Hours]

[Maximum Marks : 75

Before answering the questions, candidates should ensure that they have been supplied the correct and complete question paper. No complaint in this regard, will be entertained after examination.

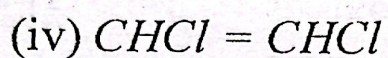
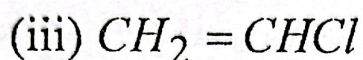
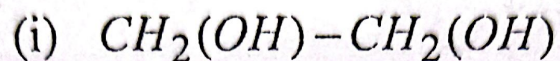
Note : Attempt *all* questions from Section-A & any *two* questions from Section-B and any *seven* questions from Section-C.

SECTION – A

1. Multiple Choice Questions. $1 \times 20 = 20$
- (a) Compounds which have different arrangements of atoms in space have :
- (i) position isomerism
 - (ii) functional group isomerism
 - (iii) chain isomerism
 - (iv) stereoisomerism

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(b) Which of the following compounds shows optical isomerism ?



(c) If a molecule has element of symmetry it is called as :

(i) Asymmetric

(ii) Symmetric

(iii) Meso compound

(iv) None

(d) Relative Configuration is denoted by :

(i) R

(ii) S

(iii) D

(iv) α

(e) Which conformation has lowest energy ?

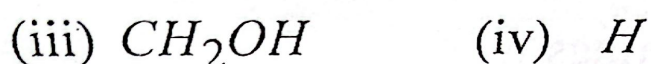
(i) Gauche

(ii) Fully Eclipsed

(iii) Staggered

(iv) Partially Eclipsed

6 (f) Which of the following groups has the lowest priority according to the Cahn-Ingold-Prelog sequence rules ?



7 (g) Stereoisomers which do not have mirror images relationship are called as :

(i) Enantiomers

(ii) Racemic Mixture

(iii) Meso Compounds

(iv) Diastereomers

(h) Cyclohexane exists in conformations :

(i) 4

(ii) 5

(iii) 6

(iv) None

(i) Cis -Trans isomerism is a part of :

(i) Tautomerism

(ii) Metamerism

(iii) Geometrical Isomerism

(iv) None

- 10 (j) Which of the following is strong hydride donor ?
- (i) NaBH_4 (ii) LiAlH_4
- (iii) B_2H_2 (iv) None of the above

- 11 (k) Which of the following is *not* true about the five membered rings ?

- (i) More stable than 4 membered rings
- (ii) More stable than 6 membered rings
- (iii) More stable than 7 membered rings
- (iv) More stable than 8 membered rings

- 12 (l) Which of the following catalyst is used for reduction of Pyridine ?

- (i) H_2/Pt
- (ii) $\text{HI}/\text{Red P}$
- (iii) LiAlH_4
- (iv) All of the above

- 13 (m) Which of the following five membered rings is most stabilized by resonance ?

- (i) Furan (ii) Thiophene
- (iii) Pyrrole (iv) Pyridine

14 (n) What is the product when thiophene reacts with Br_2 in benzene ?

- (i) 2-bromothiophene
- (ii) 3-bromothiophene
- (iii) 2,5-dibromothiophene
- (iv) 3,4-dibromothiophene

15 (o) What is the product when pyrrole reacts with Br_2 in ethanol ?

- (i) 2,3-dibromopyrrole
- (ii) 2,3,4,5-tetrabromopyrrole
- (iii) 2,5- dibromopyrrole
- (iv) 3,4-dibromopyrrole

16 (p) Electrophillic attack in Quinoline occurs at :

- (i) C5
- (ii) C4
- (iii) C4 & 8
- (iv) ~~C5~~ C5 & 8

17 (q) Which element is present as hetero atom in Thiophene ?

- (i) Sulphur
- (ii) Nitrogen
- (iii) Oxygen
- (iv) Sulphur and nitrogen

18 (r) The electron of Nitrogen participating in the resonance in pyridine is present in which orbital ?

(i) p-orbital (ii) ~~(ii)~~ sp²-orbital

(iii) sp³-orbital (iv) sp-orbital

19 (s) Pyridine is an aromatic compound :

~~(i)~~ True (ii) False

20 (t) How many numbers of resonating structures stabilises a Thiophene molecule ?

(i) 4 (ii) ~~(ii)~~ 5

(iii) 6 (iv) None

SECTION - B

2. Long Answer Questions (Attempt any *two*) : $2 \times 10 = 20$

(a) Explain Electrophillic substitution reactions of Quinoline and Indole.

(b) Explain Metal hydride and Clemmensen Reduction.

(c) What are Conformations ? Explain different conformations of Cyclohexane.

SECTION - C

3. Short Answer Questions (Attempt any *seven*) :

$$7 \times 3 = 35$$

- (a) Explain Geometrical Isomerism.
- (b) Explain Medicinal uses of Imidazole and Pyrazole.
- (c) Is Pyridine Aromatic ? Give Explanation.
- (d) Write a note on Absolute Configuration.
- (e) Explain Enantiomers and Diastereomers.
- (f) Write a note on Elements of Symmetry.)
- (g) Explain Huckels Rule.
- (h) Explain Claisen-Schmidt condensation.
- (i) Write a note on Meso compounds.

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B. Pharmacy 4th Semester 1st Sessional Examination, March 2024
PHARMACEUTICAL ORGANIC CHEMISTRY-III
Subject code: BP401T

Date: March 11, 2024

Time: 1.5 Hr

Max. Marks: 30

SECTION-A

Section-A is compulsory

(10X2)

- ✓ a. Draw structure of pyrazole and state its two medicinal uses.
- b. State the conditions of optical isomerism.
- c. Compare the aromaticity of pyrrole and furan.
- d. Define meso compounds with examples.
- e. Define enantiomers and diastereoisomers with examples

SECTION-B

Attempt any 2 questions from this section

(2X5)

2. Give RS system of nomenclature of optical isomers with sequence rules.
- ✓ 3. Give two methods of synthesis and four reactions of pyrrole.
4. What is racemic modification? Explain the methods of resolution of a racemic mixture?

SECTION-C

Attempt any one question from this section

(1X10)

5. Give three methods of synthesis and six chemical reactions of furan. ✓
6. Elaborate the synthesis and reactions of Imidazole ✓

B. Pharmacy 4th Semester 2nd Sessional Examination, April 2024
PHARMACEUTICAL ORGANIC CHEMISTRY-III (BP401T)

Date: April 22, 2024

Time: 1.5 Hr

Max. Marks: 30

Section-A is compulsory(5X2)

- a. Why is pyridine basic in nature?
- b. Write conformations of ethane.
- c. Give the reaction and mechanism of Dakin's reaction. ✓
- d. Draw the structure of any two heterocyclic compounds containing two hetero atoms.
- ✓ e. Write two methods of synthesis of acridine.

SECTION-B

Attempt any 2 questions from this section(2X5)

2. Give the reaction, mechanism and synthetic importance of Sodium borohydride reduction. ✓
- ✓ 3. Write at least 2 synthetic methods for each of the pyrimidines and purines along with their medicinal uses.
4. Write a note on Beckmanns rearrangement. ✓

SECTION-C

Attempt any one question from this section

(1X10)

5. Give the synthesis, reactions and medicinal uses of quinoline. ✓
6. Define geometrical isomerism with example. Write 8 different methods used for determination of configuration of geometrical isomers with examples.